

Supplementary Slides

Dealing with high complexity in spatio-temporal models

Mario Figueira Pereira

January 21, 2026

PhD program in *Statistics and Optimization*
Department of Statistics and Operations Research
University of Valencia

Table of contents

- 1 Bayesian Feedback

- 2 Sequential Consensus

- 3 Distributed & Recursive inference

- 4 Future research lines

1. Bayesian Feedback

Transferring strategies in *Bayesian Feedback*

In the *Bayesian Feedback* procedure we can explore different strategies to perform the transfer of marginal information:

- **Full prior update:** In this approach, we employ the default Bayesian learning, replacing the prior distribution with the posterior distribution. While this is a straightforward process for marginal hyperparameter distributions using the **table** feature, defining the latent field as Gaussian is necessary.
 - ▶ Marginal update
 - ▶ Joint update
- **Prior update by moments:** In this alternative approach, the prior distribution is approximated by updating the characteristic moments based on the posterior distribution.
 - ▶ Same kernels
 - ▶ Different kernels

However, we only apply this procedure to fixed effects and hyperparameters.

Bayesian Feedback diagrams (I)

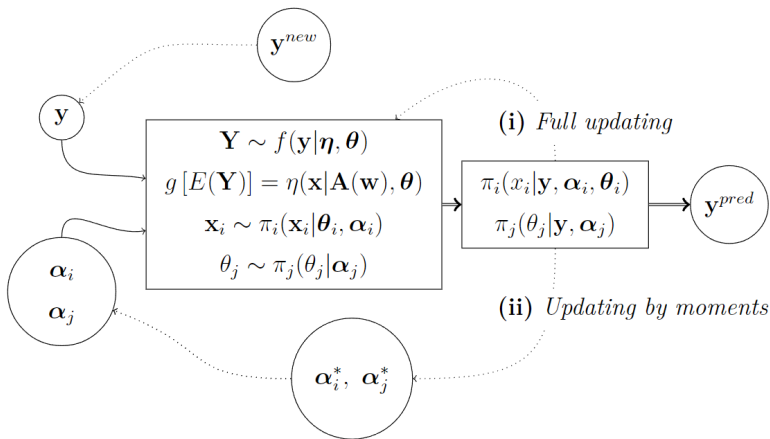


Figure 1: Bayesian feedback scheme, illustrating the transfer of information through the marginal distributions of the model's parameters.

Bayesian Feedback diagrams (II)

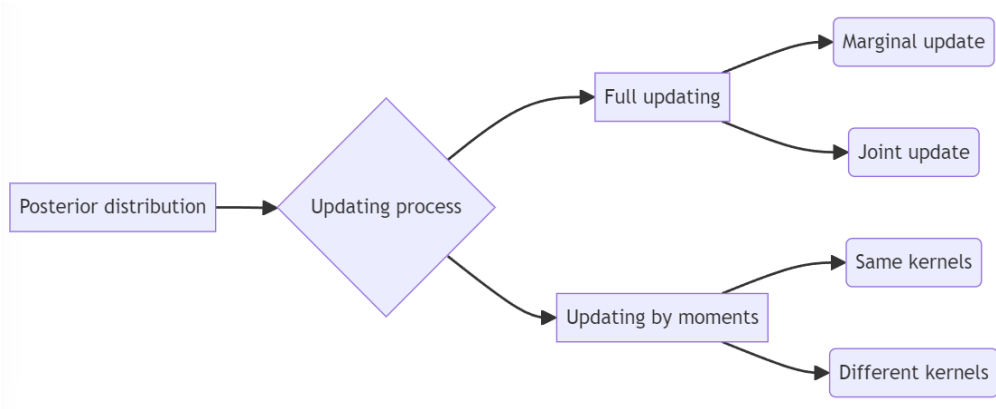


Figure 2: *Bayesian feedback* diagram for the different strategies to defined and transfer marginal information.

2. Sequential Consensus

Sequential Consensus

The information-transfer procedure for the Sequential Consensus (SC) and Sequential Consensus for Partitions (SCP) for algorithms implied the following step (●-SC & ●-SCP):

1. (●-●) Perform a *Bayesian Feedback* procedure for the sequence of the data $\mathbf{y} = \{\mathbf{y}_i, \dots, \mathbf{y}_n\}$.

$$\begin{aligned}\pi(X_k \mid \mathbf{y}_{1:i}) &\longrightarrow \pi(X_k \mid \mathbf{y}_{1:(i+1)}) \propto \pi(\mathbf{y}_{i+1} \mid X_k) \pi(X_k \mid \mathbf{y}_{1:i}), \\ \pi(\theta_m \mid \mathbf{y}_{1:i}) &\longrightarrow \pi(\theta_m \mid \mathbf{y}_{1:(i+1)}) \propto \pi(\mathbf{y}_{i+1} \mid \theta_m) \pi(\theta_m \mid \mathbf{y}_{1:i}).\end{aligned}$$

2. (●) Consensus of the random effects.
3. (●) Fixing the CCD, or other integration scheme from the last step of the *Bayesian Feedback* procedure, $\{\boldsymbol{\theta}^k\}_{k=1}^K$.
4. (●) Refit the model for each partition $\mathbf{y}_i$ by redefining the prior distributions of the fixed effects $\pi(\boldsymbol{\beta} \mid \mathbf{y}_{-i})$, leveraging the fixed support points $\{\boldsymbol{\theta}^k\}_{k=1}^K$ through the **user integration points** option of the **R-INLA** package.
5. (●) Consensus of the random effects.

2. Sequential Consensus

2.1. Removing information in the SCP

Removing information in the SCP

The information control implies that, to compute the proper marginal posterior of the random effects of the latent field ($\mathbf{x}_r$), we need to use as prior distribution of the fixed effects for every partition the marginal given the complementary data $\pi(\boldsymbol{\beta} \mid \mathbf{y}_{-i})$. We estimate $\pi(\boldsymbol{\beta} \mid \mathbf{y}_{-i})$ as:

$$\pi(\boldsymbol{\beta} \mid \mathbf{y}) \propto \left[\prod_{j=1}^i \pi(\mathbf{y}_j \mid \boldsymbol{\beta}) \pi(\boldsymbol{\beta}) \right] \times \prod_{k=i+1}^n \pi(\mathbf{y}_k \mid \boldsymbol{\beta}) \Rightarrow \tilde{\pi}(\boldsymbol{\beta} \mid \mathbf{y}_{-i}) = \frac{\pi(\boldsymbol{\beta} \mid \mathbf{y}) \pi(\boldsymbol{\beta} \mid \mathbf{y}_{1:(i-1)})}{\pi(\boldsymbol{\beta} \mid \mathbf{y}_{1:i})}$$

If we assume that $\pi(\boldsymbol{\beta} \mid \mathbf{y})$, $\pi(\boldsymbol{\beta} \mid \mathbf{y}_{1:i})$ and $\pi(\boldsymbol{\beta} \mid \mathbf{y}_{1:(i-1)})$ are Gaussians then $\pi(\boldsymbol{\beta} \mid \mathbf{y}_{-i})$ is also Gaussian, with mean (μ^*) and precision (τ^*) computed as:

$$\begin{aligned} \mu^* &= \mu_n + \mu_{i-1} - \mu_i, \\ \tau^* &= (\tau_n + \tau_{i-1} - \tau_i) \times (\tau_n \mu_n + \tau_{i-1} \mu_{i-1} - \tau_i \mu_i). \end{aligned}$$

3. Distributed & Recursive inference

A “numerical” example for beginners

Let's consider an extremely simple case: a Gaussian model with an intercept and fixed variance: $y_i \sim \mathcal{N}(\mu_i, \sigma^2)$ and $\mu_i = \beta$, where $\beta \sim \mathcal{N}(0, \tau = 0.001)$.

In this scenario, if σ is fixed and the data is split into an arbitrary partition $\mathbf{y} = \mathbf{y}_1, \dots, \mathbf{y}_n$, the posterior distribution becomes:

$$\pi(\beta \mid \mathbf{y}) = \frac{\pi(\mathbf{y} \mid \beta)\pi(\beta)}{\pi(\mathbf{y})} \propto \left[\prod_{i=1}^n \pi(\mathbf{y}_i \mid \beta) \right] \pi(\beta) = \left[\prod_{i=1}^n \pi(\mathbf{y}_i \mid \beta) \pi(\beta)^{1/n} \right],$$

which, in the log-scale, is expressed as: $\log \pi(\beta \mid \mathbf{y}) \propto \sum_{i=1}^n [\log \pi(\mathbf{y}_i \mid \beta) + \frac{1}{n} \log \pi(\beta)]$.

Therefore, the parallel result aligns with the inference obtained from using the entire dataset:

$$\log \pi(\beta \mid \mathbf{y}) \propto - \sum_{i=1}^n \left[\frac{(\mathbf{y}_i - \beta)^2}{2\sigma^2} + \frac{1}{n} \frac{\beta^2}{2 \cdot 10^3} \right] = - \sum_{i=1}^n \frac{(\mathbf{y}_i - \beta)^2}{2\sigma^2} - \frac{\beta^2}{2 \cdot 10^3}.$$

Latent field without hyperparameters

We can now examine the scenario where we have a model with a latent field but no hyperparameters (by fixing them):

- consider a model $\mathcal{M} \equiv \{\pi(\mathbf{y} \mid \mathbf{A}(\mathbf{w})\mathbf{x}), \mathbf{A}(\mathbf{w}), \pi(\mathbf{x})\}$ without hyperparameters,
- assume that we have an arbitrary partition $\mathbf{y} = \{\mathbf{y}_1, \dots, \mathbf{y}_n\}$ and
- use the conditional independence $\pi(y_i \mid \eta_i, \mathbf{y}_{-i}) = \pi(y_i \mid \eta_i)$.

Then, we can recover the joint posterior distribution of the latent field performing a parallel inference as:

$$\begin{aligned}\pi(\mathbf{x} \mid \mathbf{y}) &\propto \pi(\mathbf{y} \mid \mathbf{A}(\mathbf{w})\mathbf{x})\pi(\mathbf{x}) \\ &= \prod_{i=1}^n [\pi(\mathbf{y}_i \mid \mathbf{A}(\mathbf{w})\mathbf{x})\pi(\mathbf{x})^{1/n}].\end{aligned}$$

Therefore, in the log-scale $\log \pi(\mathbf{x} \mid \mathbf{y}) \propto \sum_{i=1}^n [\log \pi(\mathbf{y}_i \mid \mathbf{A}(\mathbf{w})\mathbf{x}) + \frac{1}{n} \log \pi(\mathbf{x})]$, such that $\log \pi(\mathbf{x}) \propto \log |\mathbf{Q}| - \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x}$.

Recalling recursive inference

The sequential update of information can be conceptualized as a **recursive Bayesian inferential process**. In such a case, the global dataset is (naturally or artificially) divided into several subsets $\mathbf{y} = \mathbf{y}_1, \dots, \mathbf{y}_n$. In a simple example where the global dataset is divided into only two subsets ($\mathbf{y} = \mathbf{y}_1, \mathbf{y}_2$), the joint posterior distribution would be as follows:

$$\begin{aligned}\pi(\mathbf{x}, \boldsymbol{\theta} | \mathbf{y}) &\propto \overbrace{\pi(\mathbf{y}_2 | \mathbf{x}, \boldsymbol{\theta}, \mathbf{y}_1)}^{\text{Data distribution}} \cdot \overbrace{\pi(\mathbf{x}, \boldsymbol{\theta} | \mathbf{y}_1)}^{\text{Full update}}, \\ &\propto \pi(\mathbf{y}_2 | \mathbf{x}, \boldsymbol{\theta}, \mathbf{y}_1) \cdot \pi(\mathbf{y}_1 | \mathbf{x}, \boldsymbol{\theta}) \cdot \pi(\mathbf{x}, \boldsymbol{\theta}).\end{aligned}$$

The main issue lies in the distribution of the subset data, which are conditioned on the previous subsets $\pi(\mathbf{y}_2 | \mathbf{x}, \boldsymbol{\theta}, \mathbf{y}_1)$. However, if we assume conditional independence $\{\pi(y_i | \mathbf{x}, \boldsymbol{\theta}, \mathbf{y}_{-i}) = \pi(y_i | \mathbf{x}, \boldsymbol{\theta})\}$ given the latent field and hyperparameters $\{\pi(y_i | \mathbf{x}, \boldsymbol{\theta}, \mathbf{y}_{-i}) = \pi(y_i | \mathbf{x}, \boldsymbol{\theta})\}$ we would have that the expression reduces to

$$\pi(\mathbf{x}, \boldsymbol{\theta} | \mathbf{y}) \propto \pi(\mathbf{y}_2 | \mathbf{x}, \boldsymbol{\theta}) \cdot \pi(\mathbf{x}, \boldsymbol{\theta} | \mathbf{y}_1) = \pi(\mathbf{y}_2 | \mathbf{x}, \boldsymbol{\theta}) \cdot \pi(\mathbf{x} | \boldsymbol{\theta}, \mathbf{y}_1) \cdot \pi(\boldsymbol{\theta} | \mathbf{y}_1).$$

3. Distributed & Recursive inference

3.1. Computing the joint marginal posterior of the hyperparameters

Computing the joint marginal posterior of the hyperparameters (I)

The formula to compute the posterior (and the configuration) of the joint marginal posterior is, as stated before,

$$\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{y}) = \left. \frac{\pi(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta})\pi(\mathbf{x} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta})}{\tilde{\pi}_G(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{y})} \right|_{\mathbf{x}=\mathbf{x}^*(\boldsymbol{\theta})}.$$

And, in a distributed or recursive framework we computed the joint marginal as,

$$\tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}) \propto \prod_{i=1}^n \left[\frac{\pi(\mathbf{y}_i \mid \mathbf{x}, \boldsymbol{\theta})\pi(\mathbf{x} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta})}{\tilde{\pi}_G(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{y}_i)} \right]_{\mathbf{x}=\mathbf{x}_i^*} = \prod_{i=1}^n \tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{y}_i).$$

Next, to obtain the joint marginal distribution of the hyperparameters, we must first compute a Gaussian approximation to the true conditional posterior of the latent field, denoted by $\tilde{\pi}_G(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{y})$.

Computing the joint marginal posterior of the hyperparameters (II)

In distributed or recursive inference settings, however, this approximated conditional posterior may arise from two distinct sources:

- 1– the **approximated conditional posterior computed within each partition** $\tilde{\pi}_G(\mathbf{x} \mid \boldsymbol{\theta}, y_i)$,
or
- 2– the **approximation of the distributed or recursive conditional posterior** $\tilde{\pi}_{dG}(\mathbf{x} \mid \boldsymbol{\theta}, y)$.

Consequently, there are two distinct strategies for approximating and computing the joint marginal distribution of the hyperparameters.

$$\tilde{\pi}_{1d}(\boldsymbol{\theta} \mid y) \propto \prod_{i=1}^n \left[\frac{\pi(y_i \mid \mathbf{x}, \boldsymbol{\theta})\pi(\mathbf{x} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta})}{\tilde{\pi}_{dG}(\mathbf{x} \mid \boldsymbol{\theta}, y_i)} \right]_{\mathbf{x}=\mathbf{x}_i^*}, \quad \text{or} \quad \tilde{\pi}_{2d}(\boldsymbol{\theta} \mid y) \propto \left[\frac{(\prod_{i=1}^n \pi(y_i \mid \mathbf{x}, \boldsymbol{\theta}))\pi(\mathbf{x} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta})}{\tilde{\pi}_{dG}(\mathbf{x} \mid \boldsymbol{\theta}, y)} \right]_{\mathbf{x}=\mathbf{x}^*}.$$

3. Distributed & Recursive inference

3.2. Constraints in the latent field & some other details

Constraints in the latent field

In hierarchical Bayesian models, components of the latent field often encode explicit dependence structures, such as spatial or spatio-temporal effects, or correlations among fixed effects. As a result, constraints are typically required for two main reasons: to ensure identifiability—most commonly by fixing or constraining the global mean (intercept)—and to address potential impropriety of prior distributions, which could otherwise lead to ill-behaved posterior distributions.

Therefore, constraints must be imposed on the latent field in order to obtain a corrected conditional posterior, $\tilde{\pi}_G(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{y}, \mathbf{C}\mathbf{x} = \mathbf{e})$:

$$\begin{aligned}\mathbf{Q}_c &= \mathbf{Q}^{-1} - \mathbf{Q}^{-1}\mathbf{C}^\top(\mathbf{C}\mathbf{Q}^{-1}\mathbf{C}^\top)^{-1}\mathbf{C}^\top\mathbf{Q}^{-1} \\ \mathbf{x}_c &= \mathbf{x} - \mathbf{Q}^{-1}\mathbf{C}^\top(\mathbf{C}\mathbf{Q}^{-1}\mathbf{C}^\top)^{-1}(\mathbf{C}^\top\mathbf{x} - \mathbf{e})\end{aligned}$$

From this point, two possible pathways—generally leading to slightly different solutions—arise in the distributed and recursive framework: imposing the constraints either on the final distributed approximation, $\tilde{\pi}_{dG}(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{y})$, or separately within each partition, $\tilde{\pi}_G(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{y}_i)$.

Variational Bayes correction

As indicated above, the most recommended option—offering an excellent balance between accuracy and computational efficiency—is to obtain the marginal of the conditional posterior distribution $\pi(\mathbf{x}_i | \boldsymbol{\theta}, \mathbf{y})$ by marginalizing the Gaussian approximation of the conditional posterior, with a variational Bayes correction applied to the mean:

$$\begin{aligned}\mathbf{x}_{VBC} &= \mathbf{x}_0 + \mathbf{Q}_I^{-1} \tilde{\boldsymbol{\lambda}}, \\ \tilde{\boldsymbol{\lambda}} &= \arg \min_{\boldsymbol{\lambda}} \left[\mathbb{E}_{\mathbf{x} \sim \mathcal{N}(\mathbf{x} + \mathbf{Q}_I^{-1} \boldsymbol{\lambda}, \mathbf{Q}_0^{-1})} [-\log \pi(\mathbf{y} | \mathbf{x})] + \text{KLD}(\phi(\mathbf{x} | \mathbf{x}_0 + \mathbf{Q}_I^{-1} \boldsymbol{\lambda}, \mathbf{Q}_0^{-1}) || \pi(\mathbf{x})) \right].\end{aligned}$$

However, when applying the VB correction, the imposition of constraints again leads to two possible solutions: applying the correction within each partition, or applying it to the distributed or recursive conditional posterior.

3. Distributed & Recursive inference

3.3. Combined distributed and recursive framework

Combined distributed and recursive framework

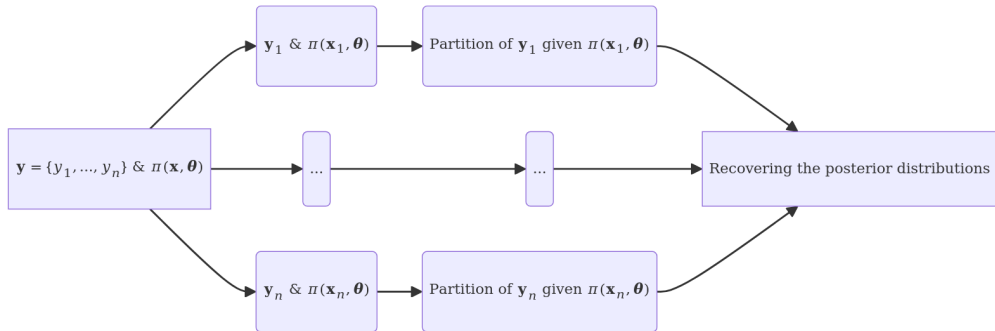


Figure 3: Scheme combining DI and RI procedures.

3. Distributed & Recursive inference

3.4. Example (DI & RI): geostatistical model

Example (DI & RI): geostatistical model (I)

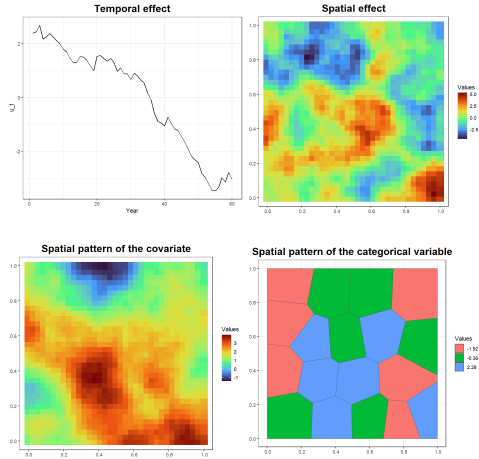


Figure 4: Simulated effects and component models.

Example (DI & RI): geostatistical model (II)

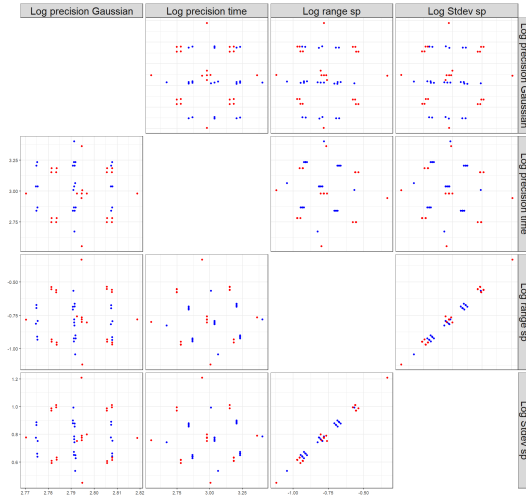


Figure 5: Central Composite Design.

Example (DI & RI): geostatistical model (III)

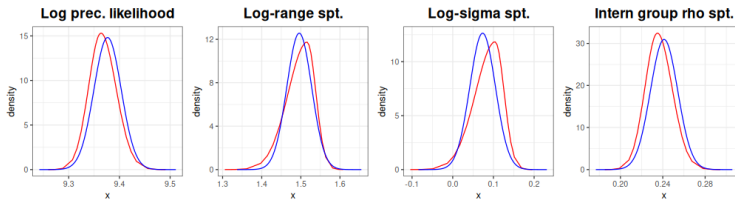


Figure 6: Marginal posterior distributions of the hyperparameters.

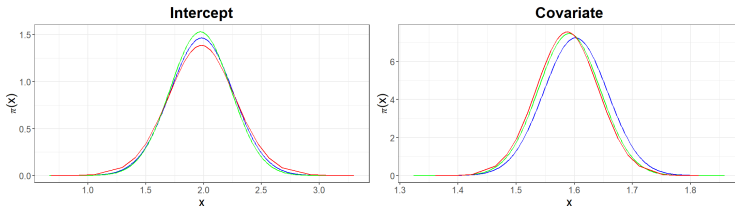


Figure 7: Marginal posterior of the fixed effects.

Example (DI & RI): geostatistical model (IV)

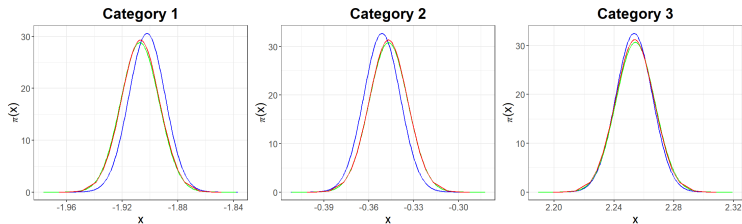


Figure 8: Marginal posterior distributions of the .

Example (DI & RI): geostatistical model (V)

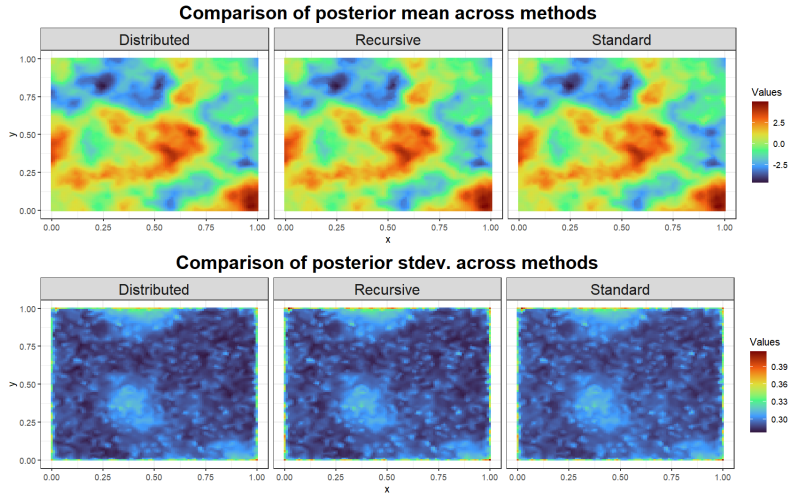


Figure 9: Mean and stdev. of the marginal posterior distribution of the spatial effect.

3. Distributed & Recursive inference

3.5. Example (DI): point process

Example (DI): point process (I)

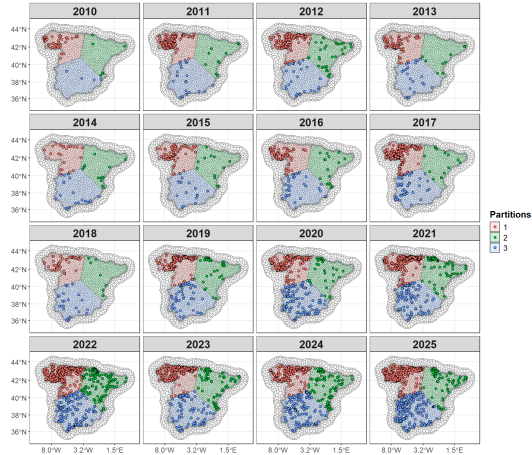


Figure 10: Point pattern data along year with the 3 partitions.

Example (DI): point process (II)

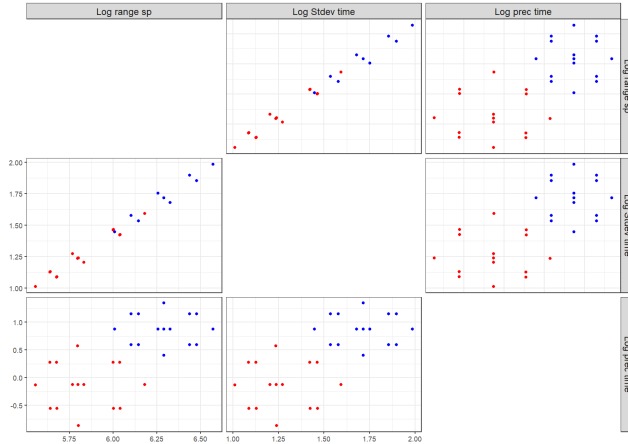


Figure 11: Central Composite Design.

Example (DI): point process (III)

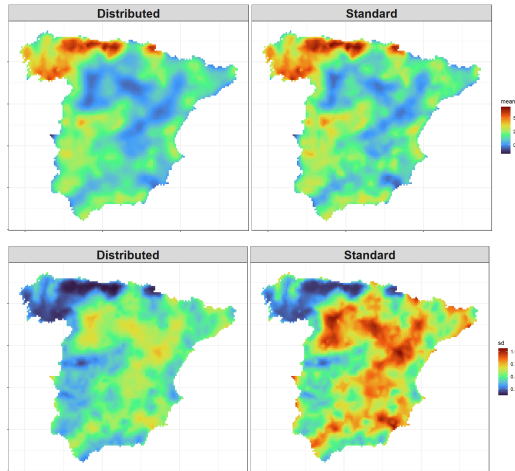


Figure 12: Spatial posterior mean (upper) and standard deviation (bottom), comparing the result from the distributed framework (left) with the standard INLA analysis (right).

Example (DI): point process (IV)

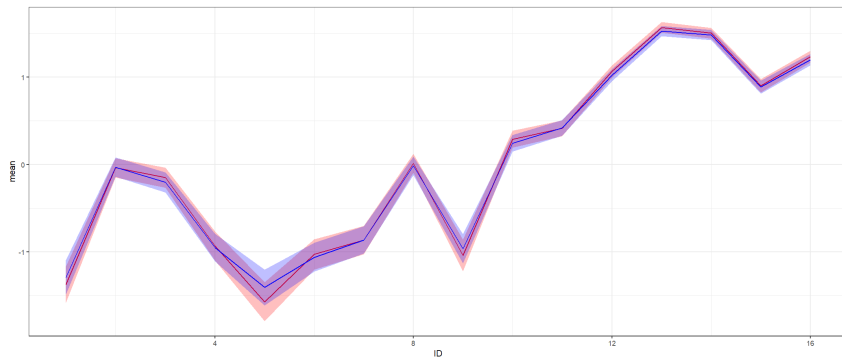


Figure 13: Posterior mean and credible interval (95%) for the temporal component.

Example (DI): point process (V)

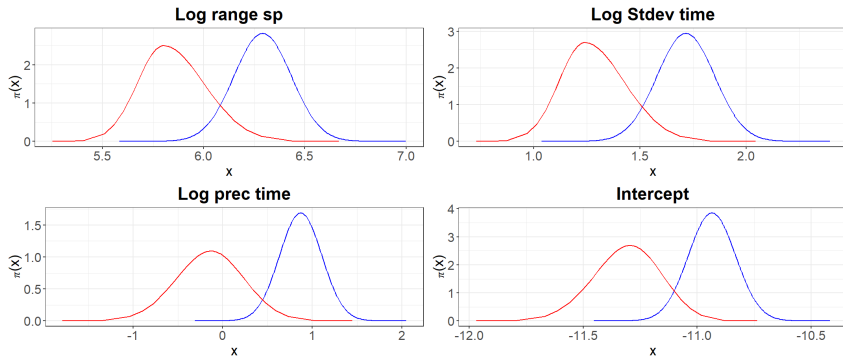


Figure 14: Marginal posterior distributions of the hyperparameters and intercept.

3. Distributed & Recursive inference

3.6. Example (RI): geostatistical model

Example (RI): geostatistical model (I)

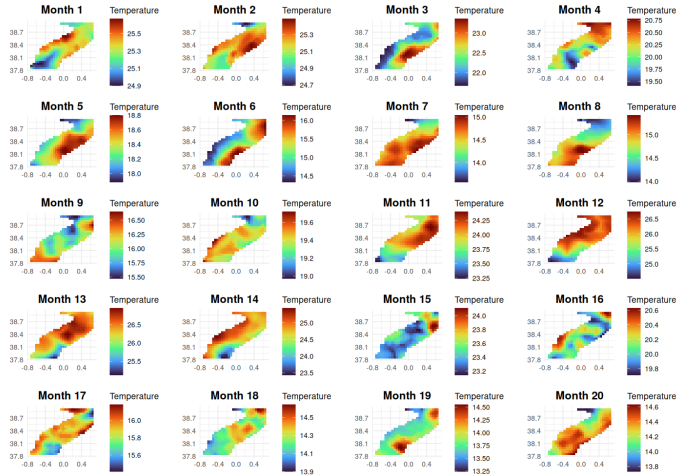


Figure 15: First 20 temporal nodes of the response variable.

Example (RI): geostatistical model (II)

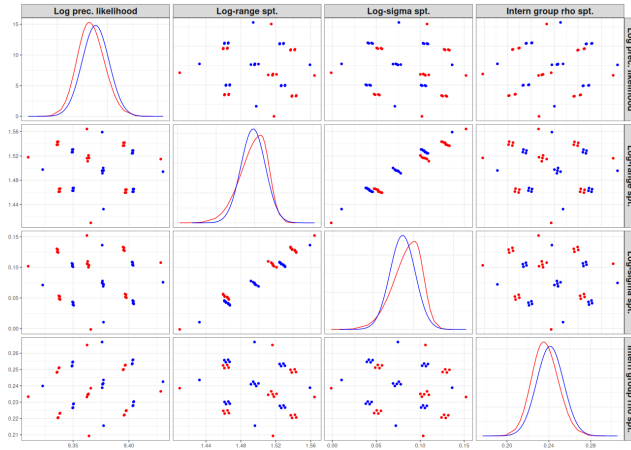


Figure 16: Central Composite Design and marginals of the hyperparameters.

Example (RI): geostatistical model (III)

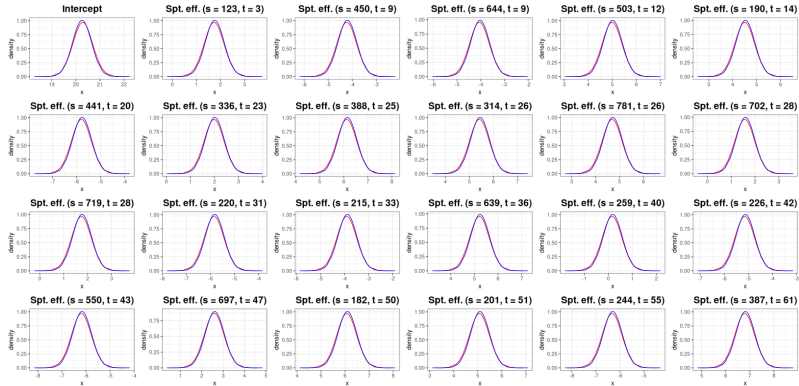


Figure 17: Some marginal distributions of the spatio-temporal effect and the intercept marginal.

4. Future research lines

Improving marginals of hyperparameters for DI and RI (I)

In the INLA approach the default strategy to compute the marginals of the hyperparameters is leveraging an assymetric Gaussian approximation:

$$\tilde{\pi}(\theta_j | \mathbf{y}) \propto \begin{cases} \exp\left(-\frac{1}{2(\sigma_j^+)^2}\theta_j^2\right) \\ \exp\left(-\frac{1}{2(\sigma_j^-)^2}\theta_j^2\right) \end{cases},$$

which leverages the following lemma to estimate the $(\sigma_j^+)^2$ and $(\sigma_j^-)^2$ paramters:

$$-\frac{1}{2}(\theta_j, \mathbb{E}(\boldsymbol{\theta}_{-j} | \theta_j))^\top \boldsymbol{\Sigma}_{\boldsymbol{\theta}}^{-1}(\theta_j, \mathbb{E}(\boldsymbol{\theta}_{-j} | \theta_j)) = -\frac{1}{2} \frac{\theta_j^2}{\tilde{\Sigma}_{jj}}.$$

By default, we compute the marginal correction for the initial Gaussian approximation of the joint marginal posterior separately for each partition, $\tilde{\pi}_G(\boldsymbol{\theta} | \mathbf{y}_i) \rightarrow \tilde{\pi}_{GMC}(\boldsymbol{\theta} | \mathbf{y}_i)$. We can then improve the estimation of the distributed marginal posterior $\tilde{\pi}_d(\theta_j | \mathbf{y})$ by exploiting the following relationship:

$$\tilde{\pi}(\boldsymbol{\theta}) = \prod_{i=1}^n \tilde{\pi}_{GMC}(\boldsymbol{\theta} | \mathbf{y}_i).$$

Improving marginals of hyperparameters for DI and RI (II)

This combination can be efficiently carried out using consensus Monte Carlo. In particular, for each Monte Carlo draw $g \in \{1, \dots, G\}$, the j -th hyperparameter is obtained as a weighted average of the corresponding draws from the n partitions:

$$\theta_{jg} = \frac{\sum_{i=1}^n w_{ijg} \theta_{ijg}}{\sum_{i=1}^n w_{ijg}}.$$

Finally, the empirical distribution of the G samples $\{\theta_{jg}\}_{g=1}^G$ provides an estimate of the marginal posterior distribution of the hyperparameters, allowing us to account for non-Gaussianity within both the distributed inference (DI) and recursive inference (RI) frameworks.